「Notation. Let $\alpha : I \to \mathbb{R}^3 ; t \mapsto (\alpha_1(t), \alpha_2(t), \alpha_3(t))$ be a regular Curve.

$$\frac{d\alpha}{dt} = \frac{d}{dt}\alpha(t) = \left(\frac{d}{dt}\alpha_1(t), \frac{d}{dt}\alpha_2(t), \frac{d}{dt}\alpha_3(t)\right).$$

「Re Parametrization

Given a regular Curve $\alpha : I \to \mathbb{R}^3$, generally, $|d\alpha/dt| \neq 1$.

But, Consider

$$S = S(t) = \int_{t_0}^{t} \left|\frac{d\alpha}{du}\right| du. \quad \text{Then, } \frac{ds}{dt} = \left|\frac{d\alpha}{dt}\right|$$

Now, $$\frac{d\alpha}{ds} = \frac{d\alpha}{dt} \cdot \frac{dt}{ds} = \frac{d\alpha}{dt} \cdot \left|\frac{d\alpha}{dt}\right|^{-1}.$$

「Def. Let $\alpha : I \to \mathbb{R}^3$ be an unit speed curve.

The tangent vector $T(s) = \alpha'(s)$

The normal vector $N(s) = \alpha''(s)/\kappa(s)$ where $\kappa(s) = \|\alpha''(s)\|$.

The binormal vector $B(s) = T \times N$

The torsion $\tau(s) = -\langle B'(s), N(s) \rangle$

$$\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

Prop. Let $\alpha(s)$ be a unit speed curve with $\kappa \neq 0$. The following are equivalent:

a). $\alpha$ is a plane curve.

b). The torsion is identically zero.

Proof. WLOG, suppose that $\alpha$ is a plane curve on the plane $z=0$. Then, we can write $\alpha(s) = (x(s), y(s), 0)$, for some differentiable $x, y$. Then, the tangent and normal are

$$T(s) = (x'(s), y'(s), 0) \text{ and } N(s) = (-y'(s), x'(s), 0).$$

So, the binormal is

$$B = T \times N = (0, 0, x'^2 + y'^2) = (0, 0, 1).$$

Now, the Binormal is constant, so $B' \equiv (0,0,0)$, so $\tau = \langle B', N \rangle = 0$.

Conversely, suppose that the torsion is zero. Then, since $B' = -\tau \cdot N = 0$, so the Binormal vector $B$ is constant. Now,

$$\frac{d}{ds}\langle \alpha(s) - \alpha(0), B \rangle' = \langle \alpha'(s), B \rangle + \langle \alpha(s) - \alpha(0), B' \rangle = \langle \alpha'(s), B \rangle = 0.$$

Therefore, $\langle \alpha(s) - \alpha(0), B \rangle$ is constant, but $s=0$, $\langle \alpha(s) - \alpha(0), B \rangle = 0$, so $\langle \alpha(s) - \alpha(0), B \rangle \equiv 0$. Now, $\alpha(s)$ lies in a plane containing $\alpha(0)$ perpendicular to $B$. $\boxtimes$